Object Oriented Programming (CST-004) - Unit 2

Unit 2 - Function Overloading, Constructors & Operator Overloading - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: function overloading, constructors and destructors, and operator overloading - making classes behave like built-in types.

Key ideas: Function overloading: same name, different parameter lists - the compiler picks by the arguments. **Constructors** run automatically at object creation: default, parameterised, and **copy constructor** (Class(const Class&) - initialises a new object from an existing one; know when it is called). The **destructor** (~Class()) runs when the object dies, in reverse order of construction. **Operator overloading** gives operators a meaning for your class, e.g. Complex operator+(Complex&). Remember: you cannot overload ::, .*, ., ?:, sizeof; and operators like =, (), [] must be member functions. A **friend function** is not a member but can touch private data - declare it inside the class with the friend keyword.

Exam pointers: writing operator+ for a Complex class (and operator++ pre/post) is the standard program. "Order of constructor/destructor calls" output questions and "difference between constructor and destructor" appear regularly.